

States of Matter

Grade 8 Science • Lesson 1



Learning objectives

- Describe the three states of matter
- Explain particle arrangement in each state
- Use the particle model to explain changes of state



What is matter?

- Matter is anything that has mass and takes up space
- Everything around us is made of tiny particles
- Particles are always moving
- How they move decides the state



Solids

- Particles packed closely in a fixed pattern
- Vibrate in place
- Fixed shape and volume



Liquids

- Particles close but can slide past each other
- Take the shape of the container
- Fixed volume



Gases

- Particles far apart and fast moving
- Fill any container
- Easy to compress



Compare: solid vs liquid

Solid

- Fixed shape
- Particles vibrate
- Hard to compress

Liquid

- Flows
- Particles slide
- Hard to compress



Quick quiz

Which state of matter has a fixed shape?

A) Solid

B) Liquid

C) Gas

D) Plasma



Summary

- Matter is made of particles
- Solids, liquids and gases differ in particle arrangement and movement
- Heating and cooling change states

